

Original Investigation

Effect of Vericiguat, a Soluble Guanylate Cyclase Stimulator, on Natriuretic Peptide Levels in Patients With Worsening Chronic Heart Failure and Reduced Ejection Fraction

The SOCRATES-REDUCED Randomized Trial

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IMPORTANCE Worsening chronic heart failure (HF) is a major public health problem.

OBJECTIVE To determine the optimal dose and tolerability of vericiguat, a soluble guanylate cyclase stimulator, in patients with worsening chronic HF and reduced left ventricular ejection fraction (LVEF).

DESIGN, SETTING, AND PARTICIPANTS Dose-finding phase 2 study that randomized 456 patients across Europe, North America, and Asia between November 2013 and January 2015, with follow-up ending June 2015. Patients were clinically stable with LVEF less than 45% within 4 weeks of a worsening chronic HF event, defined as worsening signs and symptoms of congestion and elevated natriuretic peptide level requiring hospitalization or outpatient intravenous diuretic.

INTERVENTIONS Placebo (n = 92) or 1 of 4 daily target doses of oral vericiguat (1.25 mg [n = 91], 2.5 mg [n = 91], 5 mg [n = 91], 10 mg [n = 91]) for 12 weeks.

MAIN OUTCOMES AND MEASURES The primary end point was change from baseline to week 12 in log-transformed level of N-terminal pro-B-type natriuretic peptide (NT-proBNP). The primary analysis specified pooled comparison of the 3 highest-dose vericiguat groups with placebo, and secondary analysis evaluated a dose-response relationship with vericiguat and the primary end point.

RESULTS Overall, 351 patients (77.0%) completed treatment with the study drug with valid 12-week NT-proBNP levels and no major protocol deviation and were eligible for primary end point evaluation. In primary analysis, change in log-transformed NT-proBNP levels from baseline to week 12 was not significantly different between the pooled vericiguat group (log-transformed: baseline, 7.969; 12 weeks, 7.567; difference, -0.402; geometric means: baseline, 2890 pg/mL; 12 weeks, 1932 pg/mL) and placebo (log-transformed: baseline, 8.283; 12 weeks, 8.002; difference, -0.280; geometric means: baseline, 3955 pg/mL; 12 weeks, 2988 pg/mL) (difference of means, -0.122; 90% CI, -0.32 to 0.07; ratio of geometric means, 0.885, 90% CI, 0.73-1.08; $P = .15$). The exploratory secondary analysis suggested a dose-response relationship whereby higher vericiguat doses were associated with greater reductions in NT-proBNP level ($P < .02$). Rates of any adverse event were 77.2% and 71.4% among the placebo and 10-mg vericiguat groups, respectively.

CONCLUSIONS AND RELEVANCE Among patients with worsening chronic HF and reduced LVEF, compared with placebo, vericiguat did not have a statistically significant effect on change in NT-proBNP level at 12 weeks but was well-tolerated. Further clinical trials of vericiguat based on the dose-response relationship in this study are needed to determine the potential role of this drug for patients with worsening chronic HF.

TRIAL REGISTRATION clinicaltrials.gov Identifier: [NCT01951625](https://clinicaltrials.gov/ct2/show/study?term=NCT01951625)

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More than 1 million hospitalizations for heart failure (HF) occur annually in the United States alone, and more than 80% of these hospitalized patients have worsening chronic HF.^{1,2} Despite an often rapid and substantial in-hospital improvement in HF signs and symptoms with standard therapy, approximately 25% of patients are rehospitalized within 30 days and 30% of patients may die within 1 year.^{3,4} Patients with worsening chronic HF who are not hospitalized, but require outpatient intravenous diuretic administration, may carry a similarly poor prognosis.⁵

Preclinical and clinical evidence supports augmentation of the nitric oxide-soluble guanylate cyclase (sGC)-cyclic guanosine monophosphate (cGMP) pathway as a potential therapeutic target for worsening chronic HF. Under physiologic conditions, the cGMP pathway is a critical regulator of myocardial energetics, cardiac performance, and endothelial function.⁶⁻⁹ In HF, increased inflammation and vascular dysfunction results in reduced nitric oxide bioavailability, with resultant decrease in downstream cGMP synthesis.⁹ This cGMP deficiency causes systemic, coronary, and renal microcirculatory dysfunction, which may lead to progressive myocardial damage and further inflammation.^{6,7,10} Derangements in cGMP signaling have been correlated with worse clinical outcomes among patients with reduced left ventricular ejection fraction (LVEF).⁶

Soluble guanylate cyclase stimulators offer a novel approach for addressing the relative cGMP deficit in worsening chronic HF. These agents modulate sGC in a nitric oxide-independent manner, mimicking nitric oxide and restoring downstream cGMP signaling. The recent availability of oral formulations of sGC stimulators affords the opportunity to test the hypothesis that enhancement of the cGMP pathway using an sGC stimulator will favorably influence the clinical course of worsening chronic HF. The Soluble Guanylate Cyclase Stimulator in Heart Failure with Reduced Ejection Fraction Study (SOCRATES-REDUCED) was designed to evaluate the effect on natriuretic peptide trajectory and tolerability of 12 weeks of treatment with 4 doses of oral vericiguat (BAY1021189), a novel once-daily sGC stimulator, in clinically stable patients with worsening chronic HF and reduced LVEF.

Methods

SOCRATES-REDUCED was a multicenter, randomized, double-blind, placebo-controlled, dose-finding trial that enrolled patients with worsening chronic HF after clinical stabilization. The study design has been previously described, and the protocol and statistical analysis plan are online (see the trial protocol in [Supplement 1](#) and statistical analysis plan in [Supplement 2](#)).¹¹ Institutional review board or ethics committee approval was obtained at each study site. All patients provided written informed consent.

Study Patients

The full inclusion and exclusion criteria of the SOCRATES-REDUCED study can be found in the trial protocol in [Supplement 1](#).¹¹ Briefly, eligible patients had LVEF less than 45% and a recent episode of worsening chronic HF defined by 3 components: (1) worsening HF symptoms requiring either hospital-

ization or outpatient administration of intravenous diuretics; (2) signs of congestion (clinical or chest radiograph findings); and (3) elevated natriuretic peptide level (N-terminal pro-B-type natriuretic peptide [NT-proBNP] ≥ 1000 pg/mL or B-type natriuretic peptide [BNP] ≥ 300 pg/mL if in sinus rhythm, or NT-proBNP ≥ 1600 pg/mL or BNP ≥ 500 pg/mL if in atrial fibrillation [BNP inclusion level criteria were different in Japan; see the eAppendix in [Supplement 3](#)]). All patients had a history of chronic HF, defined as New York Heart Association class II through IV symptoms and treatment with guideline-directed medical HF therapy for 30 days or longer before hospitalization or intravenous diuretic administration without hospitalization. Prior to randomization, patients were required to be clinically stable, defined as not having received intravenous vasodilator therapy for at least 24 hours and not having received intravenous diuretic therapy for at least 12 hours, and having systolic blood pressure 110 mm Hg or greater and less than 160 mm Hg, and heart rate 50/min or greater and less than 100/min.

Exclusion criteria included intravenous inotropes at any time between hospitalization and randomization; concurrent use of nitrates, phosphodiesterase type 5 inhibitors, or alternative sGC stimulators; acute coronary syndrome within 60 days prior to randomization; listing for cardiac transplant or anticipated placement of ventricular assist device; glomerular filtration rate less than 30 mL/min; and several specific HF etiologies, including hypertrophic cardiomyopathy, pericardial disease, infiltrative/inflammatory myocardial disease, and severe valvular disease.

Study Design

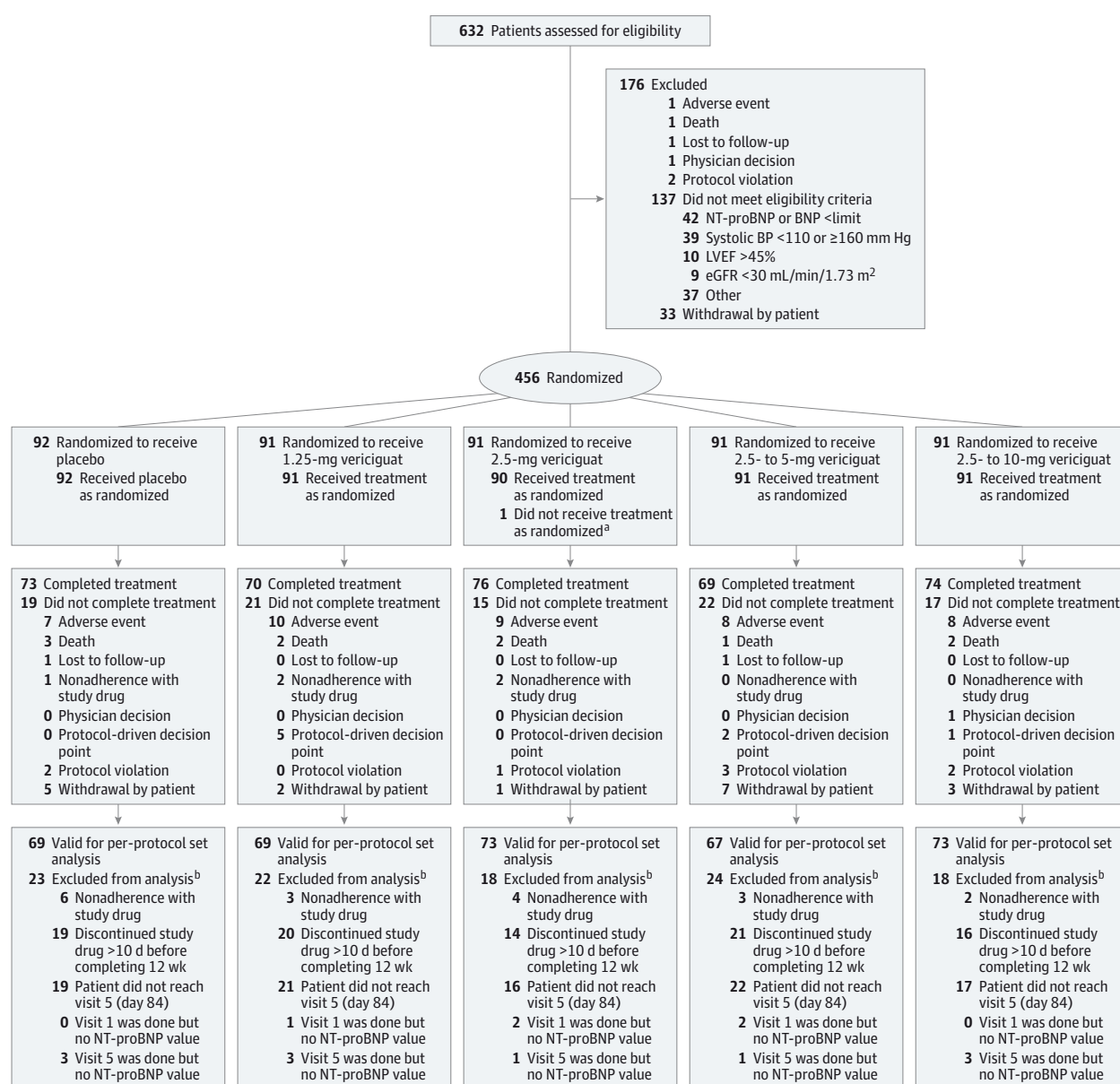
Patients who met selection criteria were randomized 1:1:1:1:1 to 1 of 5 equally sized study groups (4 vericiguat groups, 1 placebo group) ([Figure 1](#)). Patients were randomized in blocks with a block size of 10. If patient screening began during hospitalization, randomization occurred within 4 weeks of clinical stabilization, either before or after discharge. If screening began after hospital discharge, the randomization period extended up to 4 weeks after the index discharge. When screening began after outpatient intravenous diuretic therapy, randomization was permitted for 4 weeks from time of diuretic administration. Patient race/ethnicity was recorded and determined at the discretion of the investigator.

The 4 vericiguat treatment groups targeted a maximal dose of 1.25 mg, 2.5 mg, 5 mg, and 10 mg, administered orally once daily. All active treatment groups except the 1.25-mg once-daily group started with 2.5 mg once daily of vericiguat at randomization (visit 1). Study protocol specified up-titration (ie, dose doubling) or sham titration of dose at week 2 (visit 2) and week 4 (visit 3) after randomization. Dose titration in all groups was predicated on safety assessments and systolic blood pressure criteria. Planned total treatment duration was 12 weeks (visit 5), followed by a safety follow-up at 16 weeks after randomization.

Study End Points

The primary end point of the study was change from baseline to week 12 in log-transformed NT-proBNP level. All additional end points were exploratory or related to safety. Echocardiographic end points included changes in LVEF, left ventricular

Figure 1. Flow of Patients Through the Trial



NT-proBNP indicates N-terminal pro-B-type natriuretic peptide; BNP, B-type natriuretic peptide; BP, blood pressure; LVEF, left ventricular ejection fraction; eGFR, estimated glomerular filtration rate.

^a One patient in the 2.5-mg vericiguat group was too old according to local regulation (initial age limit <75 y, later suspended).

^b Major reasons for exclusion. Each patient may have >1 reason for exclusion.

end-diastolic volume (LVEDV), and left ventricular end-systolic volume (LVESV) from baseline to week 12. Additional end points included clinical events, blood pressure and heart rate, and changes in multiple biomarker levels. For the primary end point and the other biomarker end points, only blood samples analyzed by the central laboratory were considered (Covance Central Laboratory Services, Geneva, Switzerland). Echocardiograms at randomization and week 12 were read centrally by the blinded echocardiography core laboratory (Charité University Medicine, Berlin, Germany). Clinical events were adjudicated by a blinded clinical event committee. All patients, re-

searchers, and clinicians were blinded to study treatment randomization. An independent data and safety monitoring board was charged to monitor patient safety during the trial.

Statistical Analysis

A sample size of 325 patients (65 patients per study group) was estimated to provide 80% power for the primary analysis with 1-sided $\alpha = 5\%$. Assuming a 20% dropout rate, 410 was the target sample size for randomization. This calculation was based on an earlier study of an sGC stimulator in patients with reduced LVEF and pulmonary hypertension and derived from an assumed

Table 1. Baseline Characteristics (Full Analysis Set)

	Placebo (n = 92)	Vericiguat				Total (N = 456)
		1.25 mg (n = 91)	2.5 mg (n = 91)	2.5 mg to 5 mg (n = 91)	2.5 mg to 10 mg (n = 91)	
Age, mean (SD), y	67 (13)	68 (13)	68 (12)	67 (12)	69 (12)	68 (12)
Male sex, No. (%)	73 (79.3)	70 (76.9)	72 (79.1)	74 (81.3)	77 (84.6)	366 (80.3)
Weight, mean (SD), kg	78 (17)	80 (21)	81 (17)	83 (18)	81 (18)	80 (18)
Body mass index, mean (SD) ^a	27 (5)	28 (6)	28 (5)	29 (5)	28 (5)	28 (5)
Race, No. (%)						
White	69 (75.0)	68 (74.7)	73 (80.2)	75 (82.4)	76 (83.5)	361 (79.2)
Black	7 (7.6)	2 (2.2)	2 (2.2)	5 (5.5)	1 (1.1)	17 (3.7)
Asian	16 (17.4)	20 (22.0)	16 (17.6)	11 (12.1)	14 (15.4)	77 (16.9)
Other	0	1 (1.1)	0	0	0	1 (0.2)
Region, No. (%)						
Eastern Europe	19 (20.7)	20 (22.0)	27 (29.7)	23 (25.3)	24 (26.4)	113 (24.8)
Western Europe	47 (51.1)	40 (44.0)	46 (50.5)	50 (54.9)	50 (54.9)	233 (51.1)
Asia Pacific	17 (18.5)	20 (22.0)	16 (17.6)	12 (13.2)	16 (17.6)	81 (17.8)
North America	9 (9.8)	11 (12.1)	2 (2.2)	6 (6.6)	1 (1.1)	29 (6.4)
Initial worsening chronic heart failure presentation, No. (%)						
Hospitalization	71 (77.2)	72 (79.1)	76 (83.5)	68 (74.7)	68 (74.7)	355 (77.9)
Intravenous diuretic	21 (22.8)	19 (20.9)	15 (16.5)	23 (25.3)	23 (25.3)	101 (22.1)
Time from stabilization to randomization, mean (SD), d	15.0 (9.9)	14.8 (9.7)	15.0 (9.5)	14.3 (8.9)	13.0 (10.8)	14.4 (9.8)
Ischemic heart failure etiology, No. (%)	51 (55.4)	46 (50.5)	57 (62.6)	42 (46.2)	46 (50.5)	242 (53.1)
NYHA class at baseline, No. (%)						
I/II	54 (58.7)	44 (48.4)	47 (51.7)	44 (48.4)	51 (56.0)	240 (52.7)
III/IV	38 (41.3)	47 (51.6)	44 (48.4)	47 (51.7)	40 (44.0)	216 (47.3)
Systolic blood pressure, mean (SD), mm Hg	123.9 (14.3)	125.8 (13.8)	124.6 (11.8)	125.3 (12.6)	128.4 (14.6)	125.6 (13.5)
Diastolic blood pressure, mean (SD), mm Hg	75.2 (10.2)	73.9 (8.9)	73.5 (10.0)	74.6 (8.8)	75.8 (11.7)	74.6 (10.0)
Heart rate, mean (SD), /min	73.5 (13.8)	72.5 (12.6)	72.3 (11.8)	70.6 (11.6)	69.7 (11.9)	71.7 (12.4)
NT-proBNP, median (IQR), pg/mL	4043 (1962-6712)	3670 (1483-8382)	2721 (1500-8467)	2644 (1432-4687)	2805 (1526-6255)	3076 (1551-6322)
Troponin T, median (IQR), µg/L	0.026 (0.017-0.039)	0.028 (0.018-0.046)	0.025 (0.018-0.042)	0.023 (0.015-0.038)	0.027 (0.020-0.041)	0.027 (0.017-0.041)
eGFR, mean (SD), mL/min/1.73 m ²	57.8 (17.4)	57.2 (21.0)	56.9 (19.1)	60.1 (20.2)	60.0 (19.6)	58.4 (19.5)
Serum sodium, mean (SD), mEq/L	138.3 (3.4)	138.4 (3.7)	137.9 (3.2)	139.0 (2.9)	138.4 (3.6)	138.4 (3.4)
Atrial fibrillation on baseline ECG, No. (%)	30 (32.6)	32 (35.2)	30 (33.0)	30 (33.0)	32 (35.2)	154 (33.8)
Hypertension, No. (%)	70 (76.1)	71 (78.0)	70 (76.9)	68 (74.7)	78 (85.7)	357 (78.3)
Diabetes, No. (%)	41 (44.6)	36 (39.6)	54 (59.3)	39 (42.9)	49 (53.8)	219 (48.0)
Chronic kidney disease, No. (%)	38 (41.3)	35 (38.5)	41 (45.1)	37 (40.7)	35 (38.5)	186 (40.8)
Baseline echocardiographic data, mean (SD)						
LVEF, %	28.6 (8.5)	29.5 (8.6)	29.2 (8.2)	31.5 (8.5)	29.3 (8.3)	29.6 (8.4)
LVEDV, mL	174 (64)	173 (64)	174 (65)	177 (80)	161 (64)	172 (68)
LVESV, mL	127 (55)	125 (57)	126 (57)	125 (67)	120 (65)	124 (60)
Baseline therapies, No. (%)						
Diuretics	86 (93.5)	82 (90.1)	87 (95.6)	84 (92.3)	91 (100.0)	430 (94.3)
β-Blockers	83 (90.2)	77 (84.6)	79 (86.8)	86 (94.5)	86 (94.5)	411 (90.1)
ACE inhibitor	52 (56.5)	59 (64.8)	60 (65.9)	53 (58.2)	56 (61.5)	280 (61.4)
ARB	21 (22.8)	17 (18.7)	18 (19.8)	29 (31.9)	19 (20.9)	104 (22.8)
MRA	50 (54.3)	51 (56.0)	61 (67.0)	58 (63.7)	64 (70.3)	284 (62.3)
ICD	20 (21.7)	19 (20.9)	18 (19.8)	23 (25.3)	15 (16.5)	95 (20.8)
CRT-D	6 (6.5)	6 (6.6)	10 (11.0)	7 (7.7)	3 (3.3)	32 (7.0)

Abbreviations: ACE, angiotensin-converting enzyme; ARB, angiotensin II receptor blocker; CRT-D, cardiac resynchronization therapy defibrillator; ECG, electrocardiogram; eGFR, estimated glomerular filtration rate; ICD, implantable cardioverter-defibrillator; LVEDV, left ventricular end-diastolic volume; LVEF, left ventricular ejection fraction; LVESV, left

ventricular end-systolic volume; MRA, mineralocorticoid receptor antagonist; NT-proBNP, N-terminal pro-B-type natriuretic peptide; NYHA, New York Heart Association.

^a Calculated as weight in kilograms divided by height in meters squared.

difference in treatment effect of $-0.187 \log(\text{pg/mL})$ between the placebo and pooled treatment groups with a standard deviation of $0.52 \log(\text{pg/mL})$.^{11,12} The full analysis set for the present study included all patients randomized to treatment and was used to display baseline characteristics and exploratory efficacy assessments. The safety analysis set included the full analysis set of all patients who received at least 1 dose of the study drug and was used to display safety data. The per-protocol set included all patients randomized to treatment who were alive with valid measurement of NT-proBNP at baseline and week 12 and had no major protocol deviations. The per-protocol set was used for primary efficacy analysis, with the rationale being that intention-to-treat analysis would require inclusion of patients without postbaseline NT-proBNP values and thus be less appropriate for a dose-finding study. Nonetheless, several sensitivity analyses were performed including imputation by last observation carried forward, an analysis of observed cases that included patients with visit 5 NT-proBNP values and patients who dropped out due to adverse event or death, and multiple imputation analysis.

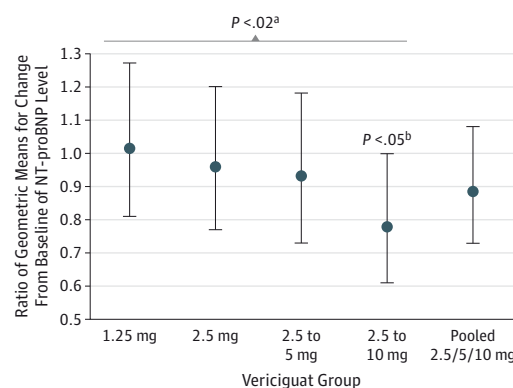
For the primary analysis of the primary end point, the 3 highest-dose vericiguat groups (2.5 mg, 5 mg, and 10 mg) were prespecified to be pooled and compared with placebo with a 2-sample *t* test at the 1-sided significance level $\alpha = 5\%$. The 1.25-mg dose group was assumed to have no or minimal effect and hence was not included in the pool of presumed effective dose groups. Secondary analyses of the primary end point were prespecified to investigate a dose-response relationship by linear regression model. In addition, if the null hypothesis of the primary analysis was rejected, pairwise comparisons of the individual vericiguat groups with placebo were planned in a hierarchical manner with a 2-sample *t* test at the 1-sided significance level $\alpha = 5\%$. If the primary analysis showed no statistical significance, comparisons of individual dose groups with placebo were to be deemed descriptive only. Only the primary and secondary analyses of NT-proBNP change from baseline to 12 weeks were prespecified in the hierarchical testing procedure and thus were α -adjusted (secondary analyses only α -adjusted if primary analysis statistically significant). All other analyses and end points were only exploratory, and hence *P* values are only descriptive and do not reflect statistical significance. SAS version 9.2 was used for all analyses.

Results

Of the 632 patients screened, 456 patients were randomized (Figure 1). The mean (SD) time from clinical stabilization to randomization was 14.4 (9.8) days. Patients were randomized from 144 sites across Europe, North America, and Asia between November 2013 and January 2015. The follow-up period ended in June 2015. Overall, 351 patients (77.0%) completed treatment with study drug with valid 12-week NT-proBNP levels and no major protocol deviation and were included in the per-protocol set for evaluation of the primary end point. There were 6 patients (1.3%) with unknown vital status at completion of safety follow-up.

The treatment groups were generally well balanced with respect to baseline characteristics, with exception of higher me-

Figure 2. Change From Baseline in NT-proBNP Level for Vericiguat Dose Groups Compared With Placebo (Per-Protocol Set)



Data are expressed as ratio of geometric means for the change from baseline to week 12 in the dosing group compared with change from baseline to week 12 in the placebo group. A value of 1.0 on the y-axis represents no difference. Number of patients included for the placebo group, 69; 1.25-mg vericiguat group, 69; 2.5-mg group, 73; 2.5- to 5-mg group, 67; 2.5- to 10-mg group, 73; and pooled 2.5/5/10-mg group, 213. Error bars indicate 90% CI; NT-proBNP, N-terminal pro-B-type natriuretic peptide.

^a Exploratory analysis of decreasing slope among vericiguat dose groups.

^b Exploratory pairwise comparison of 10-mg target dose group with placebo, 1-sided test.

dian baseline NT-proBNP level among those randomized to receive placebo and 1.25-mg vericiguat (Table 1). Overall, 77.9% of patients fulfilled worsening chronic HF criteria by being hospitalized, and 22.1% met criteria by requiring outpatient intravenous diuretic use. Mean age of the cohort was 68 years, and the majority of patients were men and white. Mean LVEF was 29.6%, and mean blood pressure was 126/75 mm Hg. Median baseline NT-proBNP level was 3076 pg/mL, and mean estimated glomerular filtration rate was 58 mL/min/1.73 m². At randomization, patients were receiving diuretics (94%), angiotensin-converting enzyme (ACE) inhibitors (61%), angiotensin II receptor blockers (23%), β -blockers (90%), and mineralocorticoid receptor antagonists (62%). The actual study medication dosing achieved per the protocol dose titration schedules is displayed in eTable 1 in Supplement 3. Of those patients randomized to receive the highest vericiguat dose and receiving study treatment on and after 8 weeks of follow-up, 71.8% reached the 10-mg daily target dose.

Primary End Point

Analyses of the primary end point are displayed in Figure 2 and Table 2. The primary end point, change in log-transformed NT-proBNP level from baseline to 12 weeks, was not significantly different in the pooled vericiguat treatment group (log-transformed values: baseline, 7.969; 12 weeks, 7.567; difference, -0.402 ; geometric means: baseline, 2890 pg/mL; 12 weeks, 1932 pg/mL) compared with the placebo group (log-transformed values: baseline, 8.283; 12 weeks, 8.002; difference, -0.280 ; geometric means: baseline, 3955 pg/mL; 12 weeks, 2988 pg/mL) (difference of means, -0.122 ; 90% CI, -0.32 to 0.07 ; ratio of geometric means on original scale, 0.885; 90% CI, 0.73-1.08; $P = .15$). Exploratory secondary analyses of the primary end point using linear regression modeling suggested a dose-response relationship ($P < .02$),

Table 2. Primary and Secondary Analyses of the Primary End Point (Change From Baseline to 12 Weeks in Log-Transformed NT-proBNP Level) (Per-Protocol Set)

Group	Baseline NT-proBNP Level (Log-Transformed [SD])	12-Week NT-proBNP Level (Log-Transformed [SD])	Change From Baseline (Ratio of Geometric Means 12-wk/Baseline, Back-Transformed)	Difference of Means (Dosing Group – Placebo, Log-Transformed), 90% CI	Ratio of Geometric Means (Dosing Group Change From Baseline/Placebo Group Change From Baseline, Back-Transformed) (90% CI)	P Value ^a
Primary Analysis of Primary End Point						
Pooled treatment groups (2.5/5/10-mg vericiguat)	7.969 (0.989)	7.567 (1.275)		−0.122 (−0.32 to 0.07)	0.885 (0.73 to 1.08)	.15
Geometric mean, back-transformed, pg/mL	2890	1932	0.669			
Placebo	8.283 (0.983)	8.002 (1.029)				
Geometric mean, back-transformed, pg/mL	3955	2988	0.755			
Secondary Analysis of Primary End Point						
2.5- to 10-mg vericiguat	8.131 (1.061)	7.602 (1.534)		−0.250 (−0.50 to 0.00)	0.779 (0.61 to 1.00)	.048
Geometric mean, back-transformed, pg/mL	3400	2001	0.589			
2.5- to 5-mg vericiguat	7.761 (0.804)	7.407 (1.156)		−0.073 (−0.31 to 0.16)	0.930 (0.73 to 1.18)	.30
Geometric mean, back-transformed, pg/mL	2346	1647	0.702			
2.5-mg vericiguat	7.998 (1.046)	7.678 (1.081)		−0.040 (−0.26 to 0.18)	0.961 (0.77 to 1.20)	.38
Geometric mean, back-transformed, pg/mL	2976	2160	0.726			
1.25-mg vericiguat	7.962 (1.188)	7.696 (1.338)		0.015 (−0.21 to 0.24)	1.015 (0.81 to 1.27)	.54
Geometric mean, back-transformed, pg/mL	2869	2201	0.767			
Secondary Analysis of Primary End Point (Linear Regression Model to Assess Dose-Response Relationship of Study Drug)						
				Intercept	Slope	P Value ^b
				−0.252	−0.026	.02

Abbreviation: NT-proBNP, N-terminal pro-B-type natriuretic peptide.

^a One-sided test.^b For decreasing slope.**Table 3. Select Clinical Exploratory End Points (Full Analysis Set)**

Clinical Outcome	Vericiguat				
	Placebo (n = 92)	1.25 mg (n = 91)	2.5 mg (n = 91)	2.5 to 5 mg (n = 91)	2.5 to 10 mg (n = 91)
Up to week 12, No. (%)					
All-cause death	3 (3.3)	4 (4.4)	3 (3.3)	3 (3.3)	2 (2.2)
CV death	3 (3.3)	4 (4.4)	2 (2.2)	2 (2.2)	2 (2.2)
HF hospitalization	16 (17.4)	16 (17.6)	16 (17.6)	9 (9.9)	9 (9.9)
CV death or HF hospitalization	18 (19.6)	17 (18.7)	18 (19.8)	11 (12.1)	10 (11.0)
End of follow-up (16 wk), No. (%)					
All-cause death	6 (6.5)	6 (6.6)	5 (5.5)	3 (3.3)	4 (4.4)
CV death	6 (6.5)	5 (5.5)	4 (4.4)	2 (2.2)	4 (4.4)

Abbreviations: CV, cardiovascular; HF, heart failure.

with higher vericiguat doses associated with greater reduction in NT-proBNP levels. Several sensitivity analyses, including imputation by last observation carried forward, an analysis of observed cases that included patients with visit 5 NT-proBNP values and patients dropping out due to adverse event or death, and multiple imputation analysis, did not meaningfully alter results. Results of sensitivity analyses for the primary and secondary

analyses of the primary end point are displayed in eTable 3, eTable 4, and eTable 5 in [Supplement 3](#).

Exploratory End Points

Pairwise exploratory analyses of individual vericiguat groups found differences in the primary end point between the 10-mg vericiguat group (log-transformed values: baseline,

Table 4. Select Echocardiographic and Vital Sign Exploratory End Points

	Change From Baseline to 12 Weeks	Difference of Means, Dosing Group – Placebo (95% CI)	P Value (2-Sided)
Echocardiographic End Points, Full Analysis Set			
LVEF, %			
Placebo	1.52		
Vericiguat			
2.5 to 10 mg	3.68	2.17 (0.33 to 4.00)	.02
2.5 to 5 mg	2.07	0.56 (–1.08 to 2.19)	.50
2.5 mg	2.74	1.23 (–0.30 to 2.76)	.12
1.25 mg	2.84	1.33 (–0.12 to 2.77)	.07
LVEDV, mL			
Placebo	–7.26		
Vericiguat			
2.5 to 10 mg	–7.32	–0.07 (–12.28 to 12.15)	.99
2.5 to 5 mg	–17.09	–9.83 (–26.01 to 6.34)	.23
2.5 mg	–9.63	–2.37 (–15.12 to 10.37)	.71
1.25 mg	–5.53	1.73 (–11.19 to 14.66)	.79
LVESV, mL			
Placebo	–6.83		
Vericiguat			
2.5 to 10 mg	–11.02	–4.19 (–14.04 to 5.67)	.40
2.5 to 5 mg	–15.49	–8.66 (–21.67 to 4.36)	.19
2.5 mg	–10.94	–4.11 (–14.14 to 5.93)	.42
1.25 mg	–8.59	–1.76 (–12.01 to 8.50)	.74
Vital Sign End Points, Safety Analysis Set			
Systolic BP, mm Hg			
Placebo	–5.14		
Vericiguat			
2.5 to 10 mg	–5.64	–0.50 (–5.14 to 4.15)	.83
2.5 to 5 mg	–3.04	2.10 (–2.69 to 6.89)	.39
2.5 mg	–3.73	1.41 (–3.40 to 6.22)	.56
1.25 mg	–4.03	1.11 (–3.21 to 5.43)	.61
Diastolic BP, mm Hg			
Placebo	–4.17		
Vericiguat			
2.5 to 10 mg	–4.05	0.13 (–3.02 to 3.28)	.94
2.5 to 5 mg	–1.34	2.84 (–0.17 to 5.84)	.07
2.5 mg	–2.94	1.24 (–2.00 to 4.47)	.45
1.25 mg	–0.49	3.69 (0.73 to 6.65)	.02
Heart rate, /min			
Placebo	–0.56		
Vericiguat			
2.5 to 10 mg	0.55	1.11 (–2.74 to 4.96)	.57
2.5 to 5 mg	–0.99	–0.43 (–4.46 to 3.60)	.83
2.5 mg	–1.56	–0.99 (–4.77 to 2.78)	.60
1.25 mg	–0.35	0.21 (–3.64 to 4.06)	.91

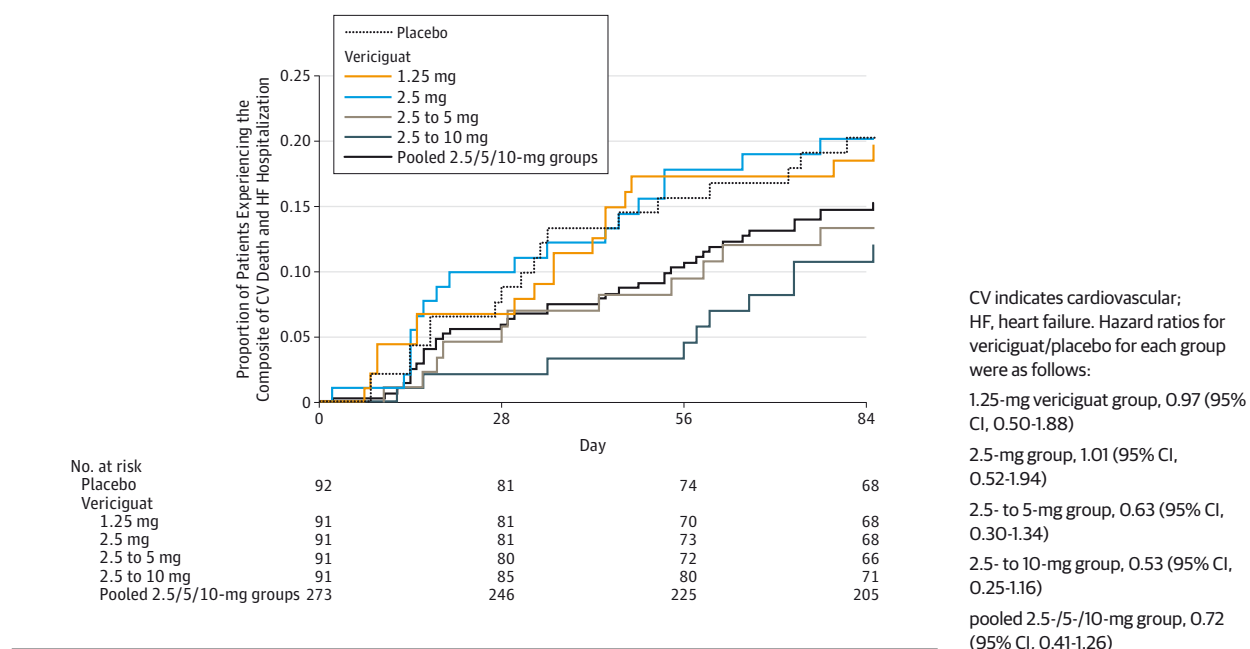
Abbreviations: BP, blood pressure; LVEDV, left ventricular end-diastolic volume; LVEF, left ventricular ejection fraction; LVESV, left ventricular end-systolic volume.

8.131; 12 weeks, 7.602; difference, –0.529; geometric means: baseline, 3400 pg/mL; 12 weeks, 2001 pg/mL) and placebo, with greater reductions in NT-proBNP level with vericiguat (difference of means, –0.250; 90% CI, –0.50 to 0.00; ratio of geometric means on original scale, 0.779; 90% CI, 0.61-1.00; $P = .048$). Comparison of the primary end point between other individual vericiguat groups and placebo all yielded $P > .05$ in these descriptive analyses.

Additional select exploratory clinical and echocardiographic end points are displayed in Table 3 and Table 4.

All-cause death as 12 weeks occurred at a rate of 3.3% among patients taking placebo and 2.2% to 4.4% among patients randomized to receive vericiguat. The rate of HF hospitalization at 12 weeks was 9.9% in the 10-mg and 5-mg vericiguat groups, 17.6% in the 2.5-mg and 1.25-mg groups, and 17.4% in the placebo group. The rate of the composite of cardiovascular death or HF hospitalization was 11% in the 10-mg vericiguat group, 12.1% in the 5-mg group, 19.8% in the 2.5-mg group, 18.7% in the 1.25-mg group, and 19.6% in the placebo group (Figure 3).

Figure 3. Time to Composite of Cardiovascular Death or Heart Failure Hospitalization



Patients in the 10-mg vericiguat group experienced an increase in LVEF at 12 weeks compared with placebo (+3.7% vs +1.5%; $P = .02$; descriptive analysis) (Table 4). There were no significant differences in change in LVEDV or LVESV between placebo and any vericiguat group. Changes in mean systolic and diastolic blood pressure and mean heart rate from baseline to 12 weeks were similar between the placebo and 10-mg vericiguat groups (all $P \geq .57$) (Table 4 and Figure 4). Data regarding changes in exploratory biomarkers are displayed in eTable 2 in Supplement 3.

Safety Outcomes

Safety outcomes were evaluated in all patients who received at least 1 dose of the study drug (455 patients, safety analysis set). Data for adverse events (AEs), serious AEs (SAEs), and study discontinuations secondary to AEs are displayed in Table 5. Rates of any AE were similar across study groups: 77.2% among patients taking placebo and as high as 78.9% among the vericiguat groups (2.5-mg group). Rates of SAE were highest among placebo patients at 39.1%. Rates of treatment-emergent acute kidney injury (placebo: 3.3%; vericiguat groups: 1.1%-5.5%) and study drug discontinuation due to SAEs (placebo: 5.4%; vericiguat groups: 2.2%-7.7%) were similar between treatment groups.

Treatment-emergent syncope was most frequent in the 10-mg vericiguat group (4 patients, 4.4%; in 3/4 patients, events occurred during the first 2 weeks after randomization, ie, when taking the 2.5-mg dose). Treatment-emergent hypotension was most common in the 10-mg vericiguat group (14 patients, 15.4%) and was next most frequent in the 2.5-mg vericiguat group (6 patients, 6.7%). Among the 14 patients with treatment-emergent hypotension in the 10-mg vericiguat group, 8 patients experienced hypotension between weeks 0 and 2, when the maximal vericiguat dose per the study titration schedule was 2.5 mg once daily. Two patients experienced hypotension between weeks

2 and 4, when the maximal vericiguat dose was 5 mg once daily. The cumulative proportion of patients with treatment-emergent hypotension during the first 2 weeks in the three 2.5-mg starting-dose groups was 4.4%, compared with 3.3% in the placebo group. Two patients in the trial discontinued the study drug due to hypotension (1 in the placebo group; 1 in the 5-mg vericiguat group).

Discussion

The SOCRATES-REDUCED trial was designed to evaluate the tolerability and optimal dose of 4 regimens of vericiguat in stabilized patients with worsening chronic HF and reduced LVEF. The primary analysis of the primary end point was not met, as pooled analysis of the 3 higher-dose vericiguat groups found no significant difference in 12-week NT-proBNP reduction compared with placebo. Prespecified secondary analyses demonstrated a dose-response relationship between increasing vericiguat dose and NT-proBNP reduction. Overall, vericiguat appeared well tolerated without an excess in AEs or SAEs compared with placebo. Further prospective testing in a phase 3 trial is needed to definitively determine the effect of vericiguat on clinical outcomes in this patient population.

The nitric oxide-sGC-cGMP pathway represents a long-established therapeutic target in HF with reduced LVEF, most notably with use of combination isosorbide dinitrate and hydralazine therapy in African American patients.¹³ More recently, targeting the particulate guanylate cyclase-cGMP pathway via neprilysin inhibition and augmentation of the natriuretic peptide system demonstrated clinical benefits.¹⁴ However, additional attempts at cGMP enhancement in HF with reduced LVEF have been challenging, as treatment with exogenous organic nitrates is limited by tolerance and impaired bioactivation.¹⁵ Moreover, oxidative stress decreases the sensitivity of sGC to nitric oxide, which

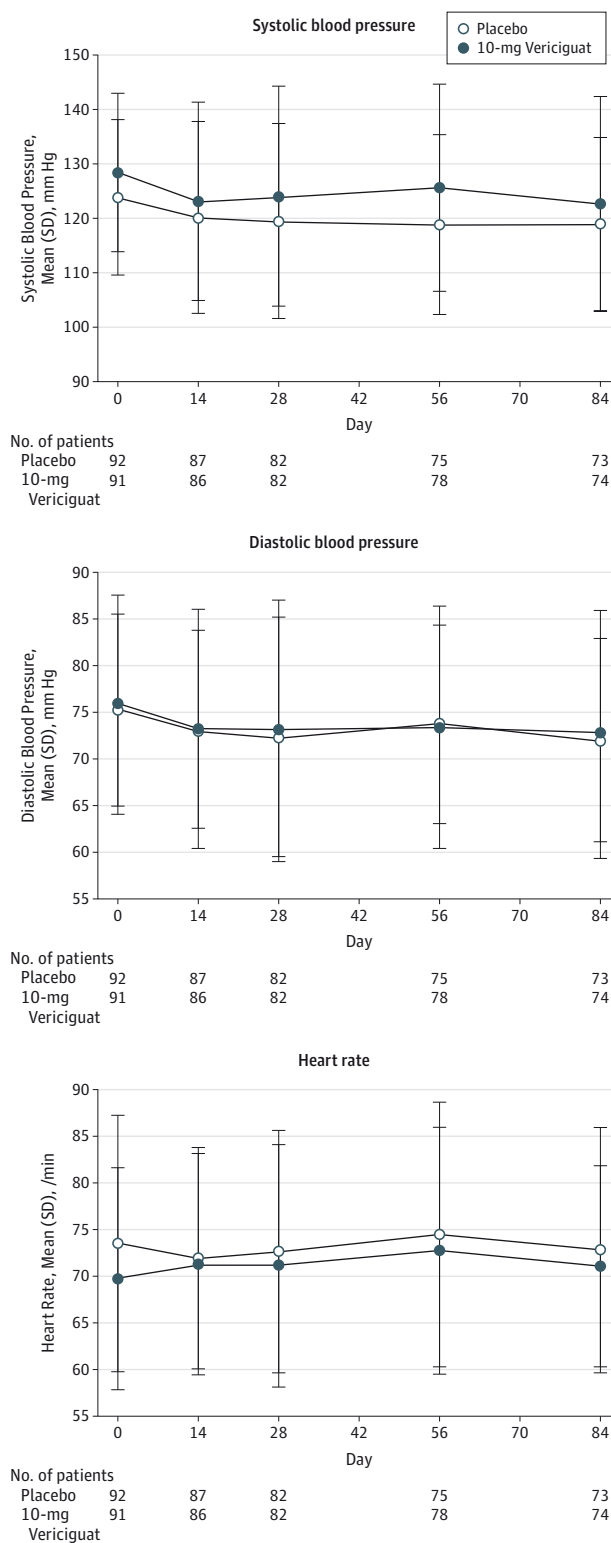
may impair downstream cGMP production even if nitric oxide levels are sufficient.⁶ Direct sGC modulators have the advantage of stimulating sGC in a nitric oxide-independent manner, circumventing problems with reduced upstream nitric oxide bioavailability and diminished endogenous activity of sGC.

Although the primary analysis of the primary end point of this trial was not met, there are several notable findings. The first is that all doses of vericiguat appeared well tolerated compared with placebo. Patients receiving placebo and those in the 2.5-mg vericiguat group had higher rates of any SAE and any AE compared with other vericiguat groups. While this study was not powered for clinical events, rates of mortality and HF hospitalization were generally numerically lower among patients receiving vericiguat, particularly among those randomized to the 2 highest-dose vericiguat groups. Specifically, the rate of HF hospitalization in these 2 highest-dose vericiguat groups was 9.9% compared with 17.4% to 17.6% in the placebo and lower-dose vericiguat groups. Although investigator-reported episodes of treatment-emergent hypotension and syncope were more frequent among patients in the highest-dose vericiguat group, these events did not prompt excess study drug discontinuation, and there were no significant differences in 12-week follow-up blood pressure and heart rate measurements among patients randomized to receive 10-mg vericiguat or placebo. Moreover, excess hypotensive events in the 10-mg vericiguat group were mostly early events (ie, ≤ 2 weeks after randomization) when patients were still receiving low-dose vericiguat identical to other study groups, thus suggesting the possibility of a chance finding.

The dose-response relationship of vericiguat with NT-proBNP level reduction in the prespecified exploratory secondary analyses warrants comment. The 10-mg vericiguat group yielded concurrent improvements in NT-proBNP level and LVEF at 12 weeks, and the observed similarities in blood pressure and heart rate between the 10-mg vericiguat and placebo groups over this period suggest that these improvements were most likely not related to reduction in diastolic blood pressure. In the setting of stable hemodynamics, these findings may signal intrinsic reverse remodeling of the heart with corresponding decreases in ventricular filling pressures and improvement in myocardial function. This conclusion would be consistent with preclinical data for sGC modulators in HF, for which these agents have been associated with reductions in cardiac fibrosis and left ventricular mass, improvement in endothelial dysfunction, and preservation of renal function, in a blood pressure-independent manner.¹⁶⁻¹⁸ In the current trial, although only exploratory in nature, the collective signal toward potential benefit in natriuretic peptide trajectory, clinical events, and LVEF at 12 weeks of treatment with the higher vericiguat dosing schedule, while maintaining safety, suggests that titration to the 5-mg or 10-mg once daily dose may be reasonable for further testing in a phase 3 study.

The contemporary treatment approach for HF with reduced LVEF centers on systemic blockade of the sympathetic and renin-angiotensin-aldosterone systems with hemodynamically active agents that decrease cardiac workload by reducing heart rate, preload, or afterload.¹⁹ These medications may lower blood pressure as an intended or unintended consequence. Continued stepwise addition of investigational agents with pronounced blood pressure-lowering effects to target doses of existing

Figure 4. Longitudinal Changes in Blood Pressure and Heart Rate



Data displayed as mean (SD) comparing the 10-mg vericiguat group with the placebo group.

guideline-directed medications raises safety and tolerability concerns.^{19,20} The present study generates the hypothesis that a once-daily sGC stimulator may exert favorable cardiac effects

Table 5. Adverse Events (Safety Analysis Set)

	No. of Patients (%)				
	Placebo (n = 92)	Vericiguat			
		1.25 mg (n = 91)	2.5 mg (n = 90)	2.5 to 5 mg (n = 91)	2.5 to 10 mg (n = 91)
Any AE	71 (77.2)	64 (70.3)	71 (78.9)	67 (73.6)	65 (71.4)
Any study drug-related AE	13 (14.1)	10 (11.0)	13 (14.4)	12 (13.2)	15 (16.5)
AE with outcome death	5 (5.4)	6 (6.6)	4 (4.4)	2 (2.2)	4 (4.4)
Any SAE	36 (39.1)	31 (34.1)	35 (38.9)	24 (26.4)	29 (31.9)
Any study drug-related SAE	3 (3.3)	1 (1.1)	1 (1.1)	1 (1.1)	4 (4.4)
Discontinuation of study drug due to AE	7 (7.6)	10 (11.0)	9 (10.0)	8 (8.8)	8 (8.8)
Discontinuation of study drug due to SAE	5 (5.4)	6 (6.6)	2 (2.2)	5 (5.5)	7 (7.7)
Treatment-emergent AEs of interest					
Hypotension ^{a,b}	6 (6.5)	5 (5.5)	6 (6.7)	4 (4.4)	14 (15.4) ^b
Asymptomatic	1 (1.1)	2 (2.2)	3 (3.3)	2 (2.2)	5 (5.5)
Symptomatic	5 (5.4)	3 (3.3)	3 (3.3)	2 (2.2)	10 (11.0)
Syncope	1 (1.1)	0	2 (2.2)	1 (1.1)	4 (4.4)
Acute kidney injury	3 (3.3)	5 (5.5)	2 (2.2)	1 (1.1)	3 (3.3)

Abbreviations: AE, adverse event; SAE, serious adverse event.

^a One patient had preferred term "orthostatic hypotension" reported.

^b One patient had both symptomatic and asymptomatic hypotension.

without sizeable changes in blood pressure and heart rate and thus may be safely tolerated for a broad HF population in combination with contemporary therapies. Although the present study required systolic blood pressure 110 mm Hg or greater for enrollment, the observation that vericiguat, as titrated in SOCRATES-REDUCED, did not meaningfully influence mean blood pressure at 12 weeks' follow-up suggests this agent could be favorable for patients with worsening chronic HF who have even lower baseline blood pressure, a subpopulation at heightened risk of adverse outcomes yet less likely to tolerate available therapies proven to reduce morbidity and mortality.²¹ Independent of hemodynamic considerations, effective therapies for worsening chronic HF and reduced LVEF are needed, and SOCRATES-REDUCED is among the first studies to test a medication for this indication.

Limitations of the current study should be acknowledged. Consistent with the dose-finding objective of this exploratory phase 2 trial, the findings from this study should be considered nonconfirmatory and descriptive, including the observed dose-response relationship in the primary end point and reduction in NT-proBNP level in the 10-mg vericiguat target dose group compared with placebo. By nature of its phase 2 design, SOCRATES-

REDUCED was not intended to make definitive claims on the clinical efficacy of vericiguat in this study population. Although change in natriuretic peptide level is a commonly used surrogate end point in HF studies due to associations with outcomes and response to proven guideline-therapies, discordance between natriuretic peptide improvement and clinical events may be seen with trials of investigational therapies.^{22,23} Moreover, although natriuretic peptide inclusion criteria differed in SOCRATES-REDUCED by presence or absence of atrial fibrillation, the effect of vericiguat on subsequent change in NT-proBNP level may have been influenced by the prevalence of atrial fibrillation in the population.²³

Conclusions

Among patients with worsening chronic HF and reduced LVEF, compared with placebo, vericiguat did not have a statistically significant effect on change in NT-proBNP level at 12 weeks but was well tolerated. Further clinical trials of vericiguat based on the dose-response relationship in this study are needed to determine the potential role of this drug for patients with worsening chronic HF.

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